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Unit – 1 : Session – 5 : SLO – 1 | Signed magnitude representation

■ Multiple Choice Questions (5)

1. d — Sign
2. b — 1101 (sign 1, magnitude 101)
3. c — Efficient hardware implementation (it is NOT a limitation — it is false; signed magnitude has complex hardware)
4. c — +2 (different signs: subtract magnitudes $5-3=2$, sign of larger = +)
5. a — Positive zero (equal magnitudes, different signs $\rightarrow 0$, conventionally +0)

■ Two-Mark Questions (2)

1. Signed magnitude is less preferred because it has two representations of zero (+0 and -0), and its arithmetic needs separate adder/subtractor hardware plus a magnitude comparator, making it slower and costlier than two's complement.
2. Rules of binary addition: $0+0=0$; $0+1=1$; $1+0=1$; $1+1=0$ with carry 1; $1+1+1=1$ with carry 1.

■ Scenario-Based Long Answer (1)

- $(-5) + (-3)$ in 4-bit signed magnitude: $-5 = 1101$, $-3 = 1011$. Same signs $\rightarrow$ add magnitudes: $101 + 011 = 1000$ (8) and keep the common sign (negative).
- But the magnitude field is only 3 bits (max 7). 8 needs 4 bits (1000) $\rightarrow$ the sum overflows the magnitude field. Result: OVERFLOW — the true result -8 cannot be represented in 4-bit signed magnitude (range -7 to +7).

■ Match the Following (5)

- A. MSB in signed magnitude — 3. Sign bit
- B. $1 + 1$ in binary — 4. 0 with carry 1
- C. Two's Complement — 1. Preferred representation today
- D. +0 and -0 — 2. Dual zero issue
- E. Overflow example — 5. $+5 + +3 = +8$ in 4-bit

■ Fill in the Blanks (5)

1. sign 2. 0 (with carry 1) 3. magnitudes 4. 0000 and 1000 5. two's complement

■ Scenario-Based Small Activity (1)

- $+4 = 0100$, $-2 = 1010$. Signs differ $\rightarrow$ compare magnitudes: $100 > 010$. Subtract smaller from larger: $100 - 010 = 010$ (2). Result takes the sign of the larger magnitude (positive).
- Final result: $0010 = +2$.

■ True or False (5)

1. FALSE — It has two zeros (+0, -0).
2. TRUE
3. FALSE — 1+1 gives 0 with carry 1.
4. TRUE
5. FALSE — Overflow occurs when the magnitude sum exceeds the magnitude range (e.g. 5+3 in 4-bit).